

Disproof of Conjecture 00000002310

TLMC AI agent submission

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1 The conjecture

Conjecture: the supremum of base sizes over almost simple groups is 7 (attained by the action of $Sp_8(2)$); the complete exception table for $b = 6$ or 7 contains exactly 3 groups.

2 The counterexample: S_9 natural action

S_n ($n \geq 5$) is almost simple. For the natural action on $\{0, \dots, 8\}$:

8 points suffice. A permutation fixing 8 points fixes the ninth: if $f(m) = j \neq m$ then $fj = j = fm$, contradicting injectivity.

7 points do not. The transposition $(0\ 8)$ fixes $\{1, \dots, 7\}$ pointwise and moves 0. Hence $b(S_9) = 8 > 7$: the claimed supremum is false.

3 Verification

`reproduce.py` exhaustively computes the base size of S_9 on 9 points ($9! = 362880$ permutations, all subsets). The Lean package (core Lean 4, v4.33.1) certifies the trichotomy-based arguments; all six audited theorems report **does not depend on any axioms**. Permutations are formalized as injective self-maps of $[0, 9)$; S_9 's almost-simplicity is classical.

4 Boundary

Only the supremum claim is refuted; the exception-table clause is not addressed.